Module 2: Global Navigation Satellite System

Course: Introduction to Drone Technology

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3 Global Navigation Satellite Systems

The other operational GNSS constellations are **GLONASS**, **Galileo**, and **BeiDou**. They differ mainly in satellite constellations, signal structures, and system characteristics. For example, Galileo provides unique signal authentication for civil users, BeiDou uses MEO, GEO, and IGSO satellites, while GLONASS has historically used FDMA signals (European Space Agency, n.d.-a; European Union Agency for the Space Programme, n.d.).

3.1 GPS architecture

The **Space Segment** consists of more than 24 MEO satellites at approximately 20,200 km altitude, transmitting navigation signals and orbital data (Official U.S. Government Information, n.d.).
The **Control Segment** consists of ground stations that monitor satellite orbits and clocks and upload updated navigation data (Official U.S. Government Information, n.d.).
The **User Segment** consists of civilian and military GPS receivers that use signals from at least four satellites to determine position and time (Official U.S. Government Information, n.d.).

3.2 GNSS error sources

The most dominant GNSS error sources are from **satellite infrastructure**, **atmospheric propagation** and **local signal environments**. For a standard single-frequency receiver, residual satellite clock errors and ephemeris errors each contribute approximately 2.1 m to the measurement inaccuracy.(Kaplan & Hegarty, 2006).

Atmospheric propagation causes significant signal delays before reaching the receiver. After applying broadcast models, the ionosphere contributes the largest error of about 4.0 m, while the troposphere adds approximately 0.7 m. Locally, signal multipath contributes around 1.0 m, and receiver measurement noise adds roughly 0.5 m. (Kaplan & Hegarty, 2006).

Because these individual measurement errors are statistically independent, they are combined using the Root Sum Square (RSS) method. This results in a total expected User Equivalent Range Error (UERE) of approximately 5.3 m, which is then multiplied by the Dilution of Precision (DOP) to determine the final position inaccuracy in meters (Kaplan & Hegarty, 2006).

3.3 Dilution of Precision (DOP)

GDOP (Geometric Dilution of Precision) describes the effect of satellite geometry on the uncertainty of the 3D position and receiver clock bias: $\mathrm{GDOP} = \sqrt{\mathrm{PDOP}^2 + \mathrm{TDOP}^2}$. **PDOP** (Position Dilution of Precision) describes the amplification of uncertainty for the 3D position, combining horizontal and vertical components: $\mathrm{PDOP} = \sqrt{\mathrm{HDOP}^2 + \mathrm{VDOP}^2}$. **HDOP** (Horizontal Dilution of Precision) and **VDOP** (Vertical Dilution of Precision) describe the geometric effect on the horizontal and vertical positions, respectively. VDOP is typically higher than HDOP because satellite geometry is naturally less favorable for determining height (European Space Agency, n.d.).

3.4 Real Time Kinematics GNSS

RTK GNSS achieves centimeter-level accuracy by using differential corrections from a known base station and carrier-phase measurements. The receiver can precisely measure the fractional part of the carrier phase, but it initially lacks the exact whole-cycle count between the satellite and the receiver (European Space Agency, n.d.).

Integer ambiguity is unknown number of full carrier signal cycles at the start of tracking. Resolving this ambiguity is strictly required to turn the carrier-phase measurements into a highly accurate fixed position solution (European Space Agency, n.d.).

3.5 GNSS accuracy

Standard Positioning Service (SPS) is standalone pseudorange-based positioning without differential corrections, providing an expected horizontal accuracy of $\approx$ 3–5 m (Kaplan & Hegarty, 2006, Ch. 7; U.S. Department of Defense, 2020).
Differential GPS (DGPS) is code-based positioning with corrections from a stationary base station to cancel common atmospheric and clock errors, achieving $\approx$ 1–2 m accuracy (Kaplan & Hegarty, 2006, Ch. 8).
RTK float is carrier-phase positioning where cycle ambiguities are estimated as floating-point values because they are not yet resolved, resulting in $\approx$ 0.2–1 m (decimeter-level) accuracy (Teunissen & Montenbruck, 2017).
RTK fixed is carrier-phase positioning with ambiguities resolved to exact integers, achieving high-precision accuracy of $\approx$ 1–2 cm horizontally (Teunissen & Montenbruck, 2017).

4 Coordinate systems

4.1 Universal Transverse Mercator (UTM) accuracy

One way to approach this is to define a reference position represented by latitude/longitude coordinates, convert this position to UTM coordinates, then project another reference position with a known distance (according to UTM) of 1 km, convert that position back to latitude/longitude coordinates, then calculate the distance between these same positions based on a Great Circle Distance formula (see Aviation Formulary document). This is partly implemented in the test script.

Using the same starting point makes it easier to compare results between teams. UTM distortion depends on the position, so different starting points could give different results even if everyone uses the same code.

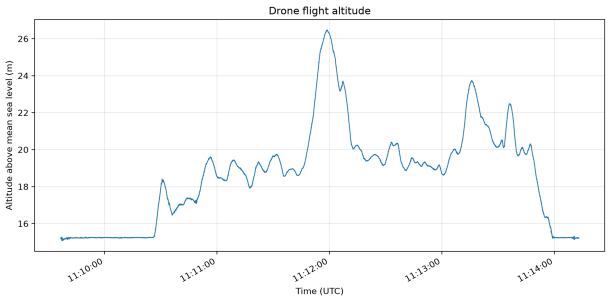
Yes. Using `utmconv`, we added 1000 m to the easting and then, in a separate calculation, to the northing. After converting back to latitude and longitude, the haversine formula gave:

Offset	UTM distance [m]	Great circle distance [m]	Difference [m]
East	1000.0000	996.9214	-3.0786
North	1000.0000	999.0788	-0.9212

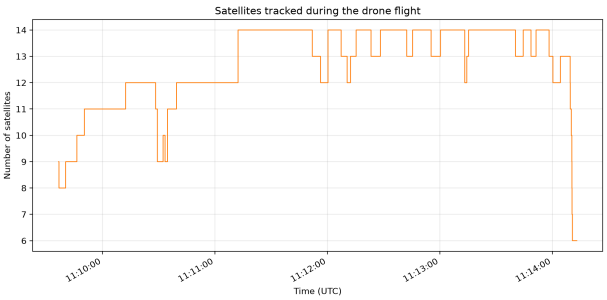
The eastward offset gives a larger difference. However, this is not just UTM distortion: the haversine formula treats Earth as a sphere, while the UTM conversion uses an ellipsoid. Locally, UTM has the same scale factor in all directions. Also, an offset along the UTM grid does not keep latitude or longitude exactly constant.

The zero keeps the coordinate format consistent. Longitude goes up to $180°$, so it is often written with three digits before the decimal point, while latitude only goes up to $90°$ and needs two. The leading zero does not change the value: `E010.33` and `E10.33` mean the same longitude.

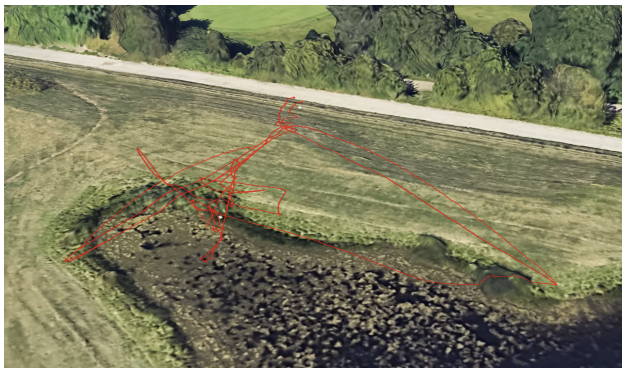
4.2 National Marine Electronics Association (NMEA) 0183 data



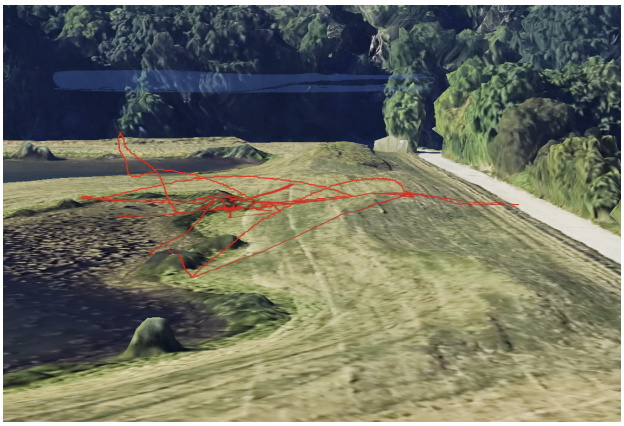
4.2.1 Altitude above Mean Sea Level.



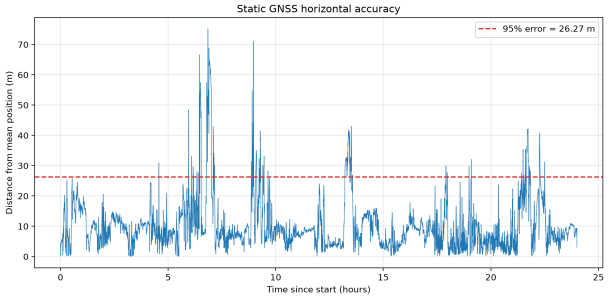
4.2.2 Satellites used during the flight.



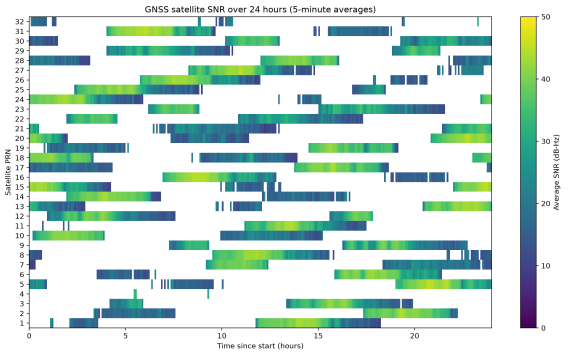
4.2.3 Drone track – angled view.



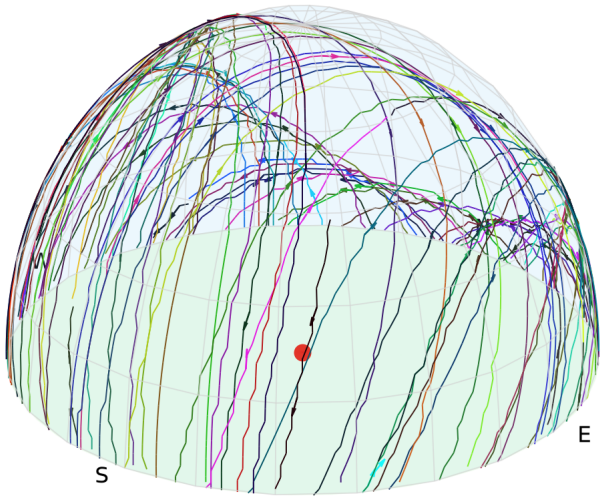
4.2.3 Drone track – ground view.



4.2.4 Static GNSS horizontal accuracy.



4.2.5 Satellite SNR over 24 hours.



4.2.6 Hemisphere map of the satellite paths of 24 hours.

4.3 Local Tangent Plane (LTP) accuracy (optional)

Offset	LTP distance [m]	Great Circle distance [m]	Difference [m]
East	1000.0000	996.6092	−3.3908
North	1000.0000	998.7666	−1.2334

The eastward offset has the larger difference because the two calculations use different models of the Earth.

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